

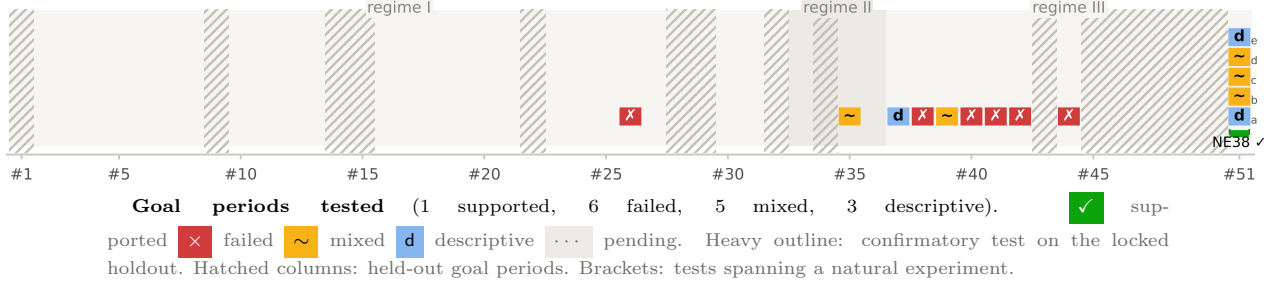
H29: Driver nodes: where an operator message moves the whole swarm

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. Content influence between agents is real and address-gated (named $\gg$ unnamed, sharper on corrected visibility); the content-pull controllability ranking is unusable, but the reply graph gives a reproducible, weakly predictive relay ranking, and operator messages stay with the agent they name.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

We built a content-influence network per goal period (how far agent j 's next statement moves toward a message from i it has just seen, net of placebos) and ranked agents by linear controllability in 9 counted and 3 short non-holdout units (regime III, #51). **Round 1b** re-ran everything on the context ledger's visibility, with a second embedding model and DQ2's reply graph. The pre-registered test still fails in the opposite direction ($\kappa < 0$ in 7/9): messages posted during the recipient's own call are co-responses, not leaked influence. Comparing seen and unseen messages at matched age, influence is real and goes through names: named recipients move 0.04–0.08 of the gap per message in every #51 segment, unnamed ones 0.00–0.01. The content-pull driver ranking is still noise, but a reply-graph ranking (net reply current) is reproducible (split-half 0.65) and weakly predicts held-out spread (pooled $\rho = 0.23$ bge, 0.12 gte). Natives: an operator's re-assignment of one agent moved it and did not relay. Holdout not run.

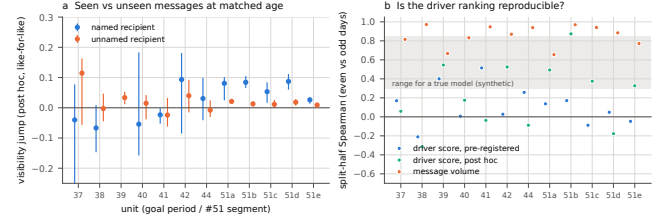
PHYSICAL MODEL

Linear response (model 02) with vector content states (11): DeGroot/Friedkin–Johnsen dynamics. x_j : agent j 's whitened statement embedding. $V(\tau)$: messages that reached j between its previous and current model calls. a_{ij} : pull per message. $A_{ij} = a_{ij}r_{ij}$, with r_{ij} the messages from i absorbed by j per hour. Γ : relaxation toward the agent's own field. Driver score D_k : reachability of the swarm mean from agent k over $T = 4$ h. $E_k = N^2/D_k$: energy to shift the mean through k .

$$x_j(\tau_n) = x_j(\tau_{n-1}) + \sum_{m \in V(\tau_n)} a_{s(m)j} [x_m - x_j(\tau_{n-1})] + \gamma_j [h_j - x_j] + \xi$$

$$\dot{\mathbf{x}} = \mathbf{Q}\mathbf{x} + \mathbf{e}_k u, \quad Q_{ji} = A_{ij}, \quad Q_{jj} = -\sum_i A_{ij} - \Gamma$$

$$D_k = \int_0^T (\mathbf{1}^\top e^{Q^t} \mathbf{e}_k)^2 dt, \quad \mathbf{1}^\top \mathbf{Q} \mathbf{e}_k = \text{out}_k - \text{in}_k - \Gamma$$



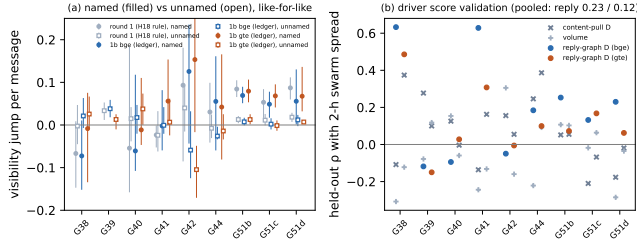
Observables. (a) Post hoc visibility jump at matched age (seen minus unseen messages within 30s of the reply), like-for-like: named recipients (blue) vs unnamed (orange). Named $\gg$ unnamed in every #51 segment; the two-room weeks are underpowered. (b) Split-half reproducibility of rankings: message volume is always reproducible; the driver score only sometimes (grey band = what a true model gives on synthetic data).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The useful part is negative and simple. To move an agent, name it: that is a 3–6 $\times$ stronger lever than broadcast, and broadcast content barely propagates. To move the whole swarm, address every agent rather than hunting for a relay. Network controllability, the hypothesis' selling point, adds nothing measurable here. In one big room the “driver” is just the chattiest agent. In the small two-room weeks the ranking is noise. And no ranking, controllability or volume, predicts observed spread. The steering-cost result (a typical agent costs $\propto N^{2.7}$, a one-room hub stays flat) is model-only. A methodological warning travels too: “seen vs unseen” placebos built from turn timing are contaminated by pauses, and content similarity is dominated by recency.

RESULTS VS PREDICTIONS

Prediction	Round 1 → 1b (ledger; bge / Verdict gte)	
S1–S4 synthetic	0/19 false positives; recovery 0.47–0.79 (unchanged)	passed / partly
P1 net pull $\kappa > 0$ in $\geq 6/9$	0/9 → 1/9 / 0/9 positive; 7/9 → 7/9 / 6/9 <i>negative</i> (co-response)	failed
P2 placebo absorbs $\geq 50\%$	median ratio 1.70 → 1.90 / 1.79	exceeded
<i>post hoc</i> matched-age jump	> 0 in 4/9 → 4/9 / 5/9; #51 named 0.05–0.09 → 0.05–0.08, unnamed 0.01–0.02 → 0.00–0.01	address-gated
P5 split-half D median ≥ 0.2	0.05 → 0.29 / 0.18 (≥ 0.4 only in G51b)	partly
P6 held-out $\rho(D, H)$ pooled ≥ 0.3	0.09 → -0.07 / 0.04 (content network)	failed
1b reply-graph driver score	split-half 0.65 / 0.66; held-out 0.23 [0.05, 0.39] / 0.12; volume ≈ 0	new, weak
P8 $E^* \propto N^{1.5-2.3}$	η 2.15 → 1.98 / 1.94 (trivially N^2 ; model-only)	not informative
P10 named $\geq 2\times$, unnamed > 0	7/9 → 7/9 / 8/9; unnamed $\kappa \leq 0$ in 7–8/9	half
Native NE38 (operator relay)	target 96th/100th pct; bystanders 71st/92nd	supported
Native #35 / #26 leaders	#35: replies $\times 1.39$, no broadcast pull; #26: no reply gain, broadcast pull rank 10/10	mixed / failed



Second figure. Round 1b. (a) Like-for-like visibility jumps per unit: named recipients (filled) keep a clear jump on the context ledger in every #51 segment, while the unnamed (open) jump shrinks toward zero; grey = round 1 (H18 rule). (b) Held-out validation: the reply-graph driver score (net reply current) predicts the 2-h swarm spread in 6–7 of 9 units; the content-pull score and message volume do not.

GOAL PERIODS IN DETAIL

G26 × failed
G35 ~ mixed
G37 d 1b: descriptive (r1 descriptive; gte descriptive)
G38 × 1b: failed (r1 failed; gte failed)
G39 ~ 1b: mixed (r1 mixed; gte mixed)
G40 × 1b: failed (r1 failed; gte mixed)
G41 × 1b: failed (r1 failed; gte failed)
G42 × 1b: failed (r1 failed; gte failed)
G44 × 1b: failed (r1 failed; gte failed)
G51a d 1b: descriptive (r1 descriptive; gte descriptive)
+5 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The positive findings are post hoc (boundary test, Amendment 2) or new in round 1b (the reply network, chosen by the re-evaluation brief). The reply-parent premium (0.11–0.19) is partly built in: DQ2 ranks parent candidates partly by content similarity, and a reply is about its parent. The held-out spread statistic replicates poorly (median 0.10 bge, 0.01 gte), so the reply ranking’s validation is weak. Named effects may partly be conversational mode. The two-room weeks have too few near-boundary rows. Natives are small: one event (NE38), six leader agent-days (#35), one short pre-period (#26).

ROUND 2

Re-freeze `confirm.py` on ledger visibility and add the reply-graph ranking as a confirmatory prediction (G47, #51 tail); test the regime-I broadcast effect of designated leaders more widely; compare the net reply current with H32’s transfer-entropy sources.

No round-2 redirects yet.